

**Exercise (4)**

(1) If  $A = \begin{bmatrix} 3 & 0 \\ 1 & k \end{bmatrix}$  and the  $\text{trace}(A) = 1$ .

- Find the value of  $k$  and the eigenvalues of  $A$ .
  - Find the eigenvalues of the matrices:  $3A$ ,  $A^T$ ,  $A^2$ ,  $A^{-1}$ .
  - Find the eigenvectors of the matrix  $A$ .
  - Verify Cayley-Hamilton theorem for the matrix  $A$  and then find  $A^{-1}$ .
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(2) If  $A = \begin{bmatrix} 3 & 1 & 1 \\ 0 & k & 1 \\ 0 & 0 & 1 \end{bmatrix}$  is singular matrix and  $k$  is constant.

- Find the value of  $k$  and the eigenvalues of  $A$ .
  - Find the eigenvalues (if possible) of the matrices  $3A$ ,  $A^T$ ,  $A^3$ ,  $A^{-1}$ .
  - Verify Cayley-Hamilton theorem for the matrix  $A$ , and use it to find  $A^{-1}$ .
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(3) If  $A = \begin{bmatrix} 1 & -3 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & k \end{bmatrix}$  and  $|A| = 6$

- Find the eigenvalues of the matrices:  $A$ ,  $2A$ ,  $A^T$ ,  $A^3$ ,  $A^{-1}$ .
  - Find the eigenvectors of the matrix  $A$ .
  - Verify Cayley-Hamilton theorem for the matrix  $A$ , and use it to find  $A^{-1}$ .
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(4) If  $A = \begin{bmatrix} k & 3 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$  and  $\text{trace}(A) = 4$

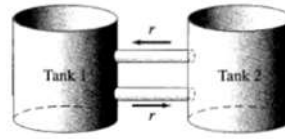
- Find the eigenvalues of the matrices  $A$ ,  $2A$ ,  $A^T$ ,  $A^3$ ,  $A^{-1}$  and  $|A|$ .
  - Find the eigenvectors of the matrix  $A$ .
  - Show that matrix  $A$  is diagonalizable and then evaluate  $A^3$  and  $e^A$ .
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(5) If  $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ , find

- Find the eigenvalues of the matrices  $A$ ,  $2A$ ,  $A^T$ ,  $A^3$ ,  $A^{-1}$  and  $|A|$ .
- a matrix  $P$  that diagonalizes the matrix  $A$  if  $A$  is diagonalizable.
- a diagonal matrix  $D$  which is similar to matrix  $A$  and then evaluate  $A^4$  and  $e^A$ .
- if  $A = PDP^{-1}$ , show that;  $A^3 = P(D^3)P^{-1}$ .

(6) Consider a closed system of two well-mixed brine tanks with volumes  $V_1 = 25$ ,  $V_2 = 40$ , liters. Mixed brine flows from tank 1 to tank 2, from tank 2 to tank 3, and from tank 3 back to tank 1 (see the opposite figure). The flow rate between each pair of tanks is  $r = 10$  liters per minute. Let  $x_1(t)$ ,  $x_2(t)$ , represent the amount of salt in tanks 1 and 2, respectively, at time  $t$  minutes. The amounts are governed by the following system of differential equations:

$$\begin{aligned} x_1'(t) &= \frac{-r}{V_1}x_1 + \frac{r}{V_2}x_2, \\ x_2'(t) &= \frac{r}{V_1}x_1 - \frac{r}{V_2}x_2, \end{aligned}$$



with initial conditions  $x_1(0) = 13$ ,  $x_2(0) = 0$ . The general solution of the above system is

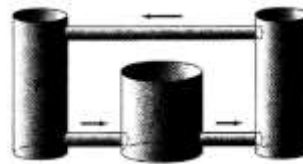
$$X(t) = c_1 e^{\lambda_1 t} V_1 + c_2 e^{\lambda_2 t} V_2,$$

where  $\lambda_1, \lambda_2$  are the eigenvalues and  $V_1, V_2$  are their associated eigenvectors, and  $c_1, c_2$  are arbitrary constants. Determine:

- The amounts of salt in each tank at time  $t$ .
- The limiting amount of salt in each tank.

(7) Consider a closed system of three well-mixed brine tanks with volumes  $V_1 = 20$ ,  $V_2 = 15$ ,  $V_3 = 4$  litres. Mixed brine flows from tank 1 to tank 2, from tank 2 to tank 3, and from tank 3 back to tank 1 (see the opposite figure). The flow rate between each pair of tanks is  $r = 60$  liters per minute. Let  $x_1(t)$ ,  $x_2(t)$ ,  $x_3(t)$  represent the amount of salt in tanks 1, 2, and 3, respectively, at time  $t$  minutes. The amounts are governed by the following system of differential equations:

$$\begin{aligned} x_1'(t) &= \frac{-r}{V_1}x_1 + \frac{r}{V_3}x_3, \\ x_2'(t) &= \frac{r}{V_1}x_1 - \frac{r}{V_2}x_2, \\ x_3'(t) &= \frac{r}{V_2}x_2 - \frac{r}{V_3}x_3. \end{aligned}$$



with initial conditions  $x_1(0) = 28$ ,  $x_2(0) = 11$ ,  $x_3(0) = 0$ . The general solution of the above system is

$$X(t) = c_1 e^{\lambda_1 t} V_1 + c_2 e^{\lambda_2 t} V_2 + c_3 e^{\lambda_3 t} V_3,$$

where  $\lambda_1, \lambda_2, \lambda_3$  are the eigenvalues and  $V_1, V_2, V_3$  are their associated eigenvectors, and  $c_1, c_2, c_3$  are arbitrary constants. Determine:

- The amounts of salt in each tank at time  $t$ .
- The limiting amount of salt in each tank.